

Rifamycins

Stephanie Sterling
Cynthia Portal-Celhay

Short View Summary

OVERVIEW

- The rifamycin class of antibiotics acts through inhibition of RNA synthesis by high-affinity binding to the DNA-dependent RNA polymerase of prokaryotes. Spontaneous resistance limits use as monotherapy. Though rifamycins differ in their pharmacokinetics and adverse reactions, most have significant drug interactions, especially with CYP3A4. In addition to the traditional uses of rifamycins in mycobacterial and staphylococcal infections, there are newer indications for both rifampin and unique rifamycins, such as rifaximin, which are used to treat gastrointestinal infections and disorders. The indications and doses of the specific drugs, administered orally unless indicated otherwise, are detailed as follows.

RIFAMPIN

- *Mycobacterium tuberculosis* infection: 600 mg daily (10 mg/kg)
- *Mycobacterium kansasii* and *Mycobacterium leprae* infection: 600 mg daily
- Staphylococcal prosthetic valve endocarditis: 300 mg three times daily (adjunct therapy)
- Staphylococcal prosthetic joint infections: 300 to 450 mg twice daily or 600 mg daily (adjunct therapy)
- Methicillin-resistant *Staphylococcus aureus* osteomyelitis in combination with trimethoprim-sulfamethoxazole (TMP-SMX), doxycycline, minocycline, clindamycin, or a fluoroquinolone based on susceptibilities: 600 to 900 mg daily (single dose or divided in two)
- Multidrug-resistant gram-negative bacilli: 10 mg/kg intravenously every 12 hours
- *Bartonella henselae* infection: 300 mg twice daily with doxycycline for complicated disease
- Brucellosis: 600 to 900 mg daily with doxycycline (second-line therapy)
- Chemoprophylaxis for infection with *Neisseria meningitidis*: 600 mg twice daily for 2 days
- Chemoprophylaxis for infection with *Haemophilus influenzae* type b: 600 mg

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RIFABUTIN

- *Mycobacterium avium-intracellulare* complex infection: 300 mg daily; 450 mg daily when used with efavirenz and 150 mg daily when used with human immunodeficiency virus (HIV) protease inhibitors
- *M. avium-intracellulare* complex prophylaxis: 300 mg daily (azithromycin is preferred)
- *M. tuberculosis*: 300 mg daily; for patients with HIV on protease inhibitors: 150 mg daily; if on efavirenz, 450 to 600 mg daily; decrease dose to 150 mg daily for creatinine clearance less than 30 mL/min

RIFAPENTINE

- Tuberculosis (active): 600 mg (10 mg/kg) twice weekly for the induction phase and once weekly for maintenance; for latent disease: 900 mg weekly with isoniazid for 12 weeks

RIFAXIMIN

- Hepatic encephalopathy: 400 mg every 8 hours; prevention: 550 mg twice daily
- Recurrent *Clostridioides difficile* (formerly *Clostridium difficile*) infection: 400 mg twice daily
- Traveler's diarrhea: 200 mg three times daily for 3 days
- Irritable bowel syndrome: 550 mg three times daily for 14 days

The rifamycins were isolated by Sensi and colleagues at the Dow-Lepetit Research Laboratories in Milan in 1959 as a mixture of five substances from fermentation cultures of the organism now classified as *Amycolatopsis mediterranei* in the Actinomycetaceae family. The nickname for the drug, "Rififi," was taken from the title of a French crime movie popular at that time and became the root of the name given to this class: the rifamycins. ¹ Rifamycin SV, the first rifamycin used clinically in 1963, was replaced by rifampicin (rifampin) in 1968 due to its improved bioavailability and greater activity against gram-positive and gram-negative bacteria and, especially, *Mycobacterium tuberculosis*. Another key role for rifampin is as an adjunct for foreign-body infections associated with biofilm production, in combination with other antibiotics for infections due to *Staphylococcus aureus* (both methicillin-susceptible *S. aureus* [MSSA] and methicillin-resistant *S. aureus* [MRSA]) and for infections caused by intracellular pathogens. ² The newer rifamycins have fewer drug interactions and improved pharmacokinetics, resulting in shorter treatment courses for tuberculosis (TB). Some of the newer rifamycins have an expanded spectrum of activity and unique pharmacokinetics and are used to treat *Clostridioides difficile* (formerly *Clostridium difficile*) infection, hepatic encephalopathy, traveler's diarrhea, and inflammatory bowel diseases. Novel investigational rifamycins and new delivery systems are in

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Structure and Mechanism of Action

The rifamycins belong to the ansamycin family (from the Latin *ansa*, meaning “handle”) of antimicrobial agents because of their basket-like structure containing an aliphatic chain connecting two ends of a naphthoquinone core. Rifampin is the 3-(4-methyl-1-piperazinyl)-iminomethyl derivative of rifamycin SV. The structures of the four rifamycins currently approved in the United States—rifampin, rifabutin, rifapentine, and rifaximin—are shown in [Fig. 26.1](#).

The antibacterial action of the rifamycins results from high-affinity binding to the DNA-dependent RNA polymerase of prokaryotes (RNAP) and inhibition of RNA synthesis. RNAP is a complex enzyme with an $\alpha_2\beta\beta'\omega$ -subunit structure. The structure of the *Thermus aquaticus* core RNAP in complex with rifampin, determined at 3.3-Å resolution, showed that rifampin binds to a site on the β -subunit deep within the DNA/RNA channel in a position that physically hinders the growing oligonucleotide chain after the first- or second-chain elongation step.³ The structure, combined with biochemical results, led to a mechanistic model for inhibition of RNAP in which rifamycins sterically block synthesis of an RNA product longer than three nucleotides (“steric-occlusion model”). Artsimovich and associates⁴ proposed a different mechanism for inhibition of RNAP by rifamycins that operates in addition to, or instead of, the steric-occlusion mechanism. Their model proposes that rifamycins also inhibit transcription at earlier steps through an allosteric signal that disfavors binding of the Mg^{2+} ion in the active site, slowing down catalysis and facilitating dissociation of short RNA transcripts. The allosteric model remains controversial because others have shown that rifamycins do not affect the affinity of binding of Mg^{3+} to the RNAP active center.⁵

Mechanisms of Resistance

Rifampin possesses highly effective bactericidal action against *M. tuberculosis* and is used as a first-line regimen against this agent. However, many exposed bacteria develop resistance to rifampin at a rate of 10^{-8} to 10^{-9} per bacterium per cell division. Mutations conferring rifampin resistance are confined almost exclusively to the *rpoB* gene in [Page 353](#) most organisms and result in a decreased affinity of the RNAP to the antibiotic. Greater than 95% of *M. tuberculosis* clinical strains resistant to rifampin harbor a mutation in an 81-base pair region of the *rpoB* gene known as the *rifampin resistance-determining region* (RRDR), which includes codons 507 to 533 and encodes 27 amino acids.⁶ Of importance, most strains have an amino acid replacement at codon 526 or have missense mutations in codon 531. Studies with patient isolates have confirmed and extended these findings.^{7, 8} Studies designed to determine the relationship between *rpoB* structural changes and rifampin minimal inhibitory concentrations (MICs) have found a strong correlation. In particular, amino acid substitutions at positions 526 and 531 conferred high-level

and extended these findings. ^{7, 8} Studies designed to determine the relationship between *rpoB* structural changes and rifampin minimal inhibitory concentrations (MICs) have found a strong correlation. In particular, amino acid substitutions at positions 526 and 531 conferred high-level resistance to rifampin, rifabutin, and rifapentine. ⁹ The molecular mechanism in the 4% of rifampin-resistant *M. tuberculosis* isolates that lack RRDR changes is unknown.

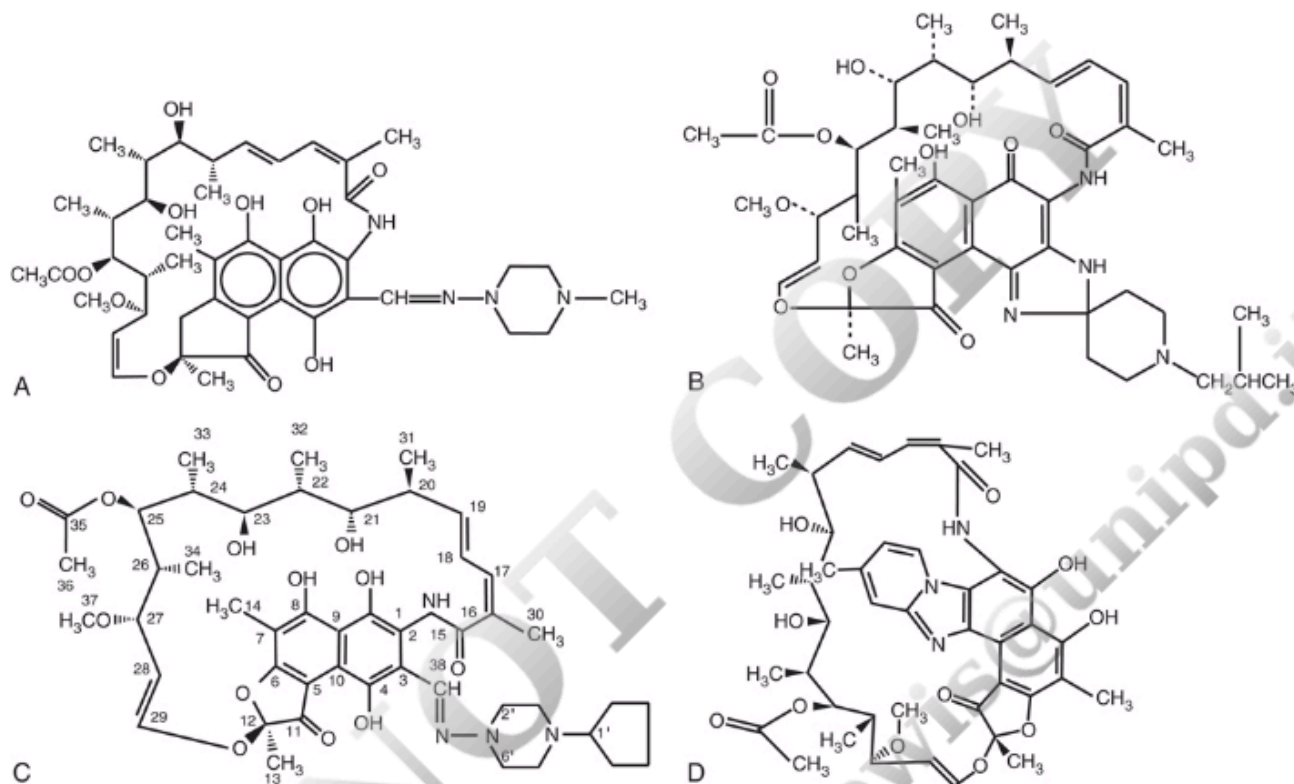


FIG. 26.1 Structures of the four major rifamycins. (A) Rifampin. (B) Rifabutin. (C) Rifapentine. (D) Rifaximin.

Follow for extended description

Alternative mechanisms of rifampin resistance also have been observed. *Pseudomonas fluorescens* has impaired cellular uptake of rifampin. Antibiotic modification occurs in strains of *Mycobacterium smegmatis* that carry the *arr* gene, causing ribosylation of rifampin and decreased susceptibility. An *arr2* gene, similar to that in *M. smegmatis*, has been identified on a plasmid in clinical isolates of Enterobacterales, including *Klebsiella pneumoniae*, and *Pseudomonas aeruginosa*. Finally, the enzyme from some species, such as *Nocardia*, is intrinsically less susceptible to rifampin inhibition because of metabolic changes at the critical hydrogen-binding sites. ¹⁰

Delineation of the molecular mechanisms of rifamycin resistance has resulted in the development and application of several polymerase chain reaction (PCR)-based strategies designed to rapidly detect mutations associated with rifampin resistance. One of the assays, Xpert MTB/RIF (Cepheid; Sunnyvale, CA), a heminested PCR, detects *M. tuberculosis* and rifampin resistance by amplifying five overlapping probes complementary to the RRDR of the *rpoB* gene and subsequently probes this region for mutations associated with rifampin resistance. ^{11, 12} Compared with conventional diagnostic methods, the Xpert MTB/RIF assay can detect *M. tuberculosis* and rifampin resistance in one sputum specimen within 2 hours; is nearly

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Other molecular assays to detect rifampin resistance are commercially available. The Genotype MTBDRplus assay (Hain Lifescience; Nehren, Germany) ¹⁶ and the INNO-LiPA Rif assay (Innogenetics; Gent, Belgium) have been validated and are currently in use in several countries. ¹⁷ In a systematic review, all three commercial tests performed well when detecting drug resistance in clinical specimens, although the Xpert MTB/RIF assay had the highest sensitivity and specificity and appeared to be the most cost-effective. ¹⁸

Although the presence of mutations in the RRDR is highly associated with rifampin resistance, false rifampin resistance detected by Xpert MTB/RIF has been reported. ¹⁹ Therefore the package insert of the Xpert MTB/RIF assay states that detection of rifampin resistance must have results confirmed by a reference laboratory. Given that the prevalence of rifampin resistance is low in the United States, the Centers for Disease Control and Prevention (CDC) recommends that a positive result indicating a mutation in the *rpoB* gene be confirmed by rapid DNA sequencing and growth-based drug susceptibility testing. ²⁰ Cases with discrepant genotypic (resistant) and phenotypic (susceptible) rifampin susceptibility results account for at least 10% of all rifampin resistance diagnoses obtained with molecular tests. Some of the discrepant strains harbor “low-level” *rpoB* mutations that confer MICs above the level determined for fully susceptible strains but below the critical concentration of 1 µg/mL currently used in standardized culture-based susceptibility tests. ²¹ The complexity of distinguishing false-positive from true-positive rifampin resistance testing results underscores the need for new guidelines to inform therapeutic decision making.

More novel diagnostic tests, such as TB intradermal reaction, ²² pyrosequencing assays, ²³ whole-genome sequencing, ²⁴ and next-generation sequencing, ^{25, 26} are being configured in a low-cost test platform to provide rapid diagnosis and drug susceptibility information for TB in the developing world and in countries dealing with a dual burden of human immunodeficiency virus (HIV) infection and multidrug-resistant (MDR) TB, where the need is acute. Page 354

TABLE 26.1

Comparative Pharmacokinetics of the Rifamycins in Healthy Adults

Drug	Bioavailability (%)	t _{max} (hr)	C _{max} (µg/mL)	t _{1/2} (hr)	Protein Binding (%)
Rifampin	68	1.5–2	8–20	2–5	80
Rifabutin	20	2.5–4	0.2–0.9	32–67	72–85 ^a

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Comparative Pharmacokinetics of the Rifamycins in Healthy Adults

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Rifabutin	20	2.5-4	0.2-0.9	32-67	72-85 ^a
Rifapentine	70 with food	4.8-6.6	8-30	14-18	98
Rifaximin	<0.4	0.8-1	3.4 ± 1.6 ng/mL	1.8-4.8	62-67.5

$C_{\max}$, Peak serum concentration; *ng*, nanograms; $t_{\max}$, time to $C_{\max}$; $t_{1/2}$, serum half-life.

^a From Blaschke TF, Skinner MH. The clinical pharmacokinetics of rifabutin. *Clin Infect Dis*. 1996;22(Suppl 1):S15-S21.

Modified from Burman WJ, Gallicano K, Peloquin C. Comparative pharmacokinetics and pharmacodynamics of the rifamycin antibacterials. *Clin Pharmacokinet*. 2001;40:327-341.

Shared Properties of the Rifamycins

The rifamycins are highly lipophilic compounds that share a common chemical structure but differ in their pharmacokinetic properties, ²⁷ summarized in **Table 26.1**. The systemic rifamycins—rifampin, rifabutin, and rifapentine—have long postantibiotic effects against *M. tuberculosis* of 68 to 75 hours. ²⁸ These drugs are metabolized in hepatocytes and intestinal microsomes to deacetylated, hydroxylated, and formyl derivatives. ²⁷ The parent drug and the active 25-*O*-deacetyl metabolite are found in serum, excreted in bile, and eliminated in feces. Repeated administration of rifampin and rifabutin results in autoinduction by gut or hepatic metabolism and increasing biliary secretion, resulting in a decrease in the area under the concentration-time curve (AUC) of both drugs and a decrease in the half-life of rifampin.

The major adverse reactions of rifamycin are divided by daily versus intermittent administration. The rifamycins have similar side effect profiles, except for rifabutin, which is discussed later. Despite the adverse reactions, the overall incidence of drug discontinuation was 1.9% in a large retrospective study, with discontinuation in over half the cases judged to be unnecessary. ²⁹

Drug-drug interactions are common with the rifamycins. The HIV integrase inhibitors, the direct-acting antivirals (DAAs) for treatment of hepatitis C virus (HCV), and ritonavir-boosted protease inhibitors for treatment of SARS-CoV-2 are recent additions to the interaction list. Coadministration of the rifamycins may lead to a decrease in the plasma concentrations of the HIV integrase inhibitors and HCV DAAs. An excellent review of rifamycin drug interactions and dosing recommendations was done by Baciewicz and colleagues. ³⁰ The antiretroviral interactions can be found in the latest US Department of Health and Human Services

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Gastrointestinal (GI) symptoms from the rifamycins are nausea, vomiting, diarrhea, and abdominal pain. Headache, fever, and pruritus can be seen. Rash may occur early and resolve without discontinuation of the drug. More severe reactions include urticarial and diffuse rash, which may be accompanied by systemic symptoms. A lupus-like syndrome with detection of antinuclear antibodies has been reported with rifampin and rifabutin, often attributed to cytochrome P (CYP) drug interactions, and usually resolves after discontinuation of the drug. ³²

Rifampin

Pharmacokinetics

Rifampin is dosed daily as 600 mg in adults and 10 to 20 mg/kg in children. The drug undergoes rapid and complete absorption, which is improved if taken on an empty stomach. Rifampin is the only rifamycin available in an intravenous formulation. It is widely distributed to most body tissues and fluids, including the cerebrospinal fluid (CSF), where doses of 20 mg/kg in children achieve mean CSF levels of 2 µg/mL. ³³ Concentrations in bone are similar to or exceed serum concentrations, ³⁴ and the drug crosses the placenta and enters breast milk. Rifampin undergoes enterohepatic circulation and deacetylation with rapid elimination in bile. Urinary excretion is 13% to 24%, with no dose adjustment necessary in renal insufficiency. A dose adjustment is recommended with severe hepatic insufficiency. The rifampin substrate is red, and the high lipophilic properties and wide distribution often turn body fluids, such as urine, tears, sweat, and feces, and contact lenses per se red-orange.

Rifampin has the greatest effect on induction of hepatic and intestinal enzymes, especially CYP3A. Induction is responsible for the majority of rifampin drug interactions. The maximal effect occurs approximately 1 week after initiation of therapy. The serum concentration of rifampin is rarely affected because the metabolism is not mediated by CYP. Most drug interactions result from induction of metabolic enzymes, especially CYP3A4, in the liver and small intestine. Rifampin and the other rifamycins have numerous interactions, outlined in **Table 26.2**, that usually result in a decrease in the AUC or maximum drug concentration (C_{max}) of the concomitant drug, requiring dose adjustments. Rifampin also induces CYP1A2, CYP2C, and CYP2D6 and some drug transporter proteins, including P-glycoprotein, a transmembrane protein that acts as Page 355 a cellular efflux pump. The HIV protease inhibitors are especially problematic when used with rifampin, with significant decreases in AUC.

TABLE 26.2

Major Rifampin Drug Interactions

CYP2D6 and some drug transporter proteins, including P-glycoprotein, a transmembrane protein that acts as [Page 355](#) a cellular efflux pump. The HIV protease inhibitors are especially problematic when used with rifampin, with significant decreases in AUC.

TABLE 26.2

Major Rifamycin Drug Interactions

Anticoagulants	Endocrine Agents
Warfarin Clopidogrel Dabigatran Rivaroxaban Apixaban	Biguanide Glucocorticoids HMG-CoA reductase inhibitors Levothyroxine Oral contraceptives Sulfonylureas Tamoxifen
Antimicrobial Agents	Neuropsychiatric and Neurologic Agents
Atovaquone Azoles Caspofungin CCR5 antagonists ^a Chloramphenicol Clarithromycin Cobicistat Dapsone Doxycycline HCV direct-acting antivirals ^d HIV integrase inhibitors ^a HIV NNRTIs ^b HIV protease inhibitors ^c Nirmatrelvir/ritonavir Linezolid Maraviroc Mefloquine Moxifloxacin Praziquantel Tenofovir alafenamide	Benzodiazepines Bupropion Citalopram Clozapine Haloperidol Lamotrigine Nortriptyline Phenytoin Risperidone
Cardiovascular Drugs	Opiates/Analgesics
β-Blockers Calcium channel blockers Digoxin Disopyramide Endothelin receptor antagonists and	Buprenorphine ^e Cyclooxygenase inhibitors Fentanyl transdermal Ketamine Methadone Oxycodone Propofol
	Other Drugs
	Cabozantinib Cediranib Deferasirox Imatinib

Praziquantel Tenofovir alafenamide Cardiovascular Drugs β-Blockers Calcium channel blockers Digoxin Disopyramide Endothelin receptor antagonists and agonists ^f Losartan Quinidine Tocainide Immunosuppressive Agents Cyclosporine Mycophenolate mofetil Sirolimus Tacrolimus	Methadone Oxycodone Propofol Other Drugs Cabozantinib Cediranib Deferasirox Imatinib Mirodenafil Omeprazole Roflumilast Theophylline Vandetanib
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HCV, Hepatitis C virus; *HIV*, human immunodeficiency virus; *HMG-CoA*, 3-hydroxy-3-methylglutaryl coenzyme A; *NNRTIs*, nonnucleoside reverse-transcriptase inhibitors.

^a Panel on Antiretroviral Guidelines for Adults and Adolescents. Guidelines for the use of antiretroviral agents in adults and adolescents living with HIV. Department of Health and Human Services; 2023. <https://clinicalinfo.hiv.gov/sites/default/files/guidelines/documents/adult-adolescent-arv/guidelines-adult-adolescent-arv.pdf> (Tables 23 and 24 a-g).

^b Efavirenz, nevirapine, etravirine, and rilpivirine.

^c Rifabutin preferred for use with HIV protease inhibitors in decreased doses.

^d Includes N53/4A protease inhibitors, nucleoside and nucleotide NS5B inhibitors, N5SA inhibitors, and nonnucleoside NS5B polymerase inhibitors.

^e Reported with rifampin only.

^f Includes bosentan, ambrisentan, and atrasentan.

Modified from Baciewicz AM, Chrisman CR, Finch CK, et al. Update on rifampin, rifabutin, and rifapentine drug interactions. *Curr Med Res Opin.* 2013;29:1-12.

Adverse Reactions

Allergic reactions to rifampin range from 0.01% to 0.26% and often occur in patients previously treated or with intermittent dosing. Type I hypersensitivity reactions, mediated by

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Adverse Reactions

Allergic reactions to rifampin range from 0.01% to 0.26% and often occur in patients previously treated or with intermittent dosing. Type I hypersensitivity reactions, mediated by immunoglobulin (Ig)E antibodies, can occur, with symptoms ranging from angioedema, bronchospasm, and urticaria to shock. Anaphylaxis can occur within minutes of the dose or be associated with prodromes, such as a rash, from prior or intermittent dosing. Patients with HIV infection had poorer outcomes, and antirifampin serum IgE antibodies were detected. ³⁵

Rifampin-associated flulike syndrome consists of fever, chills, malaise, myalgias, and headache several hours after dosing and usually resolves within several hours. Flulike symptoms are most commonly seen with intermittent regimens, although patients with HIV experience a higher incidence of flulike symptoms with higher daily dosages, which may be due to a direct toxic effect of the drug in this population. ³⁵ The incidence of flulike syndrome escalates with higher doses (900–1200 mg) given weekly and can be as high as 54%. ³⁵ The flulike syndrome usually develops after 3 months of therapy and is rarely associated with thrombocytopenia, hemolysis, and renal failure. The etiology is believed to be due to circulating IgG or IgM immune complexes that fix complement on endothelial cells or bind on the surface of cytokine-producing cells. ³⁵ After a change to weekly dosing or intermittent therapy, a more robust immune response may be seen on rechallenge. Switching to daily rifampin therapy usually results in disappearance of all symptoms. Thrombocytopenia and hemolysis are also likely immune mediated, with formation of antigen-antibody complexes that bind to erythrocytes and platelets. Recent reports highlight newly observed adverse reactions, including recurrent disseminated intravascular coagulation, with intermittent and continuous dosing. ³⁶ Acute renal failure from rifampin usually occurs with intermittent dosing and presents as acute interstitial nephritis with tubular necrosis. Less common presentations are rapidly progressive glomerulonephritis and light-chain proteinuria with polyclonal κ and λ chains. Most cases are secondary to adhesion of antirifampin antibodies to the renal tubular epithelium with subsequent complement binding. Recovery is usually complete after stopping the drug. ³⁷ Hemolysis and thrombocytopenia may occur in conjunction with renal toxicity.

Rifampin can cause changes in liver function, but serious injury is usually confined to patients with underlying liver disease due to alcohol, viral hepatitis, and other hepatotoxins, such as isoniazid, which often is coprescribed. Long-term therapy is associated with minor transient elevations in serum aminotransferase levels in 10% of patients. Liver injury from rifampin is primarily cholestatic. In most patients, serum bilirubin levels increase during the first few days of therapy and normalize spontaneously. Within several weeks of starting rifampin in patients with

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Severe hepatitis has been reported in 1.2% to 13% of patients treated with rifampin and pyrazinamide for latent TB infection, leading to updated recommendations regarding use and monitoring of this combination.⁴¹ A single nucleotide polymorphism in the gene encoding the organic anion transporting polypeptide 1B1, which plays a role in transporting bile acids and rifampin, has recently been identified with the *15 haplotype, a predisposing factor to rifampin-induced liver injury.⁴²

Antimicrobial Activity

Although commonly associated with treatment of TB, the rifamycins exhibit broad antibacterial activity. Rifampin is most potent against gram-positive bacteria, including *S. aureus*, coagulase-negative staphylococci, *Streptococcus pyogenes*, *Streptococcus pneumoniae*, viridans streptococci, *C. difficile*, and *Listeria monocytogenes*. Gram-negative bacterial coverage includes *Haemophilus influenzae*, *Neisseria meningitidis*, and *Helicobacter pylori*.⁴³ Rifampin has synergistic in vitro activity with other agents against MDR bacteria, especially *P. aeruginosa* and *Acinetobacter baumannii*. In addition to potent activity against *M. tuberculosis*, rifamycins have activity against nontuberculous mycobacteria, including *M. avium-intracellulare* complex (MAC), *M. kansasii*, and *M. marinum*.⁴⁴ Rifampin has activity against intracellular pathogens, including *Chlamydia*, *Legionella*, *Brucella*, and *Bartonella* spp.^{45, 46} The MICs of rifampin for representative organisms are shown in **Table 26.3**.

Rifabutin

Pharmacokinetics

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Rifabutin

Pharmacokinetics

Rifabutin is a spiropiperidyl rifamycin analogue with a slower and larger volume of distribution compared with rifampin and a long terminal half-life, permitting once-daily dosing. Food does not affect the AUC but does increase the peak concentration. Rifabutin is more lipid soluble than rifampin and extensively distributed throughout the body, with a CSF concentration one half that of serum after a 450-mg dose.⁴⁷ The extensive redistribution of rifabutin in tissues leads to minimum plasma concentrations of 10% to 15% of the peak 24 hours after an oral dose. The two active metabolites are 25-*O*-desacetyl rifabutin and 31-OH-rifabutin. Fifty-three percent is excreted in urine as the primary active C-25 desacetyl metabolite, and a dose reduction is recommended for a creatinine clearance less than 30 mL/min.⁴⁸ Rifabutin should be administered with caution in patients with severe hepatic disease. The ability to induce metabolic enzymes, especially CYP3A4, is least with rifabutin,²⁷ and cytochrome P4503A is the major isozyme involved in the hydroxylation of rifabutin.²⁷ When used with HIV protease inhibitors, this leads to an increase in the rifabutin level, and decreased doses are required.²⁷ Rifabutin concentrations can be significantly increased, with the potential for complex bidirectional interactions with concomitant administration of other medications, particularly in patients taking antiretrovirals. Major interactions are seen with fluconazole, ketoconazole, clarithromycin, and the HIV protease and integrase inhibitors.³¹

Adverse Reactions

Rifabutin has a unique toxicity profile, including uveitis, leukopenia, and polyarthralgias. In a study of elderly patients with pulmonary MAC treated with 600 mg of rifabutin with ethambutol, streptomycin, and a macrolide, adverse events were seen in 77%, requiring dose adjustment in over half.⁴⁹ The most common event was leukopenia, followed by GI symptoms, abnormal liver enzymes, a diffuse polyarthralgia syndrome in 19%, and anterior uveitis in 8%. Uveitis was seen in patients receiving clarithromycin and is likely secondary to a drug interaction that raises rifabutin serum concentrations. Uveitis may mimic an infectious endophthalmitis, with the presence of a hypopyon, and has been reported in patients both with and without HIV, with pulmonary MAC infection with the 600-mg dose. Cases typically occur after several months of therapy, predominantly involve the anterior chamber, and respond to corticosteroids and cycloplegics. The etiology is believed to be an immunologic reaction to cell wall proteins of dead organisms, although a direct toxic effect of rifabutin has also been suggested.⁵⁰ Uveitis has been seen in patients with HIV receiving 300-mg dosing for prophylaxis as late as 14 months after initiation,⁵¹ often in combination with fluconazole or azithromycin or both. The flulike syndrome has been reported with intermittent therapy. Hepatitis is more likely from concomitant use of hepatotoxins.

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Antimicrobial Activity

Rifabutin is active against *M. tuberculosis*, MAC, and *H. pylori*.⁵²

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TABLE 26.3

Antimicrobial Activity of Rifampin

Species	MIC ^a Range (µg/mL)	MIC ₉₀ (µg/mL)	MIC ₅₀ (µg/mL)
<i>Staphylococcus aureus</i>	0.008–0.015	0.015	0.015
<i>Staphylococcus epidermidis</i>	0.004–0.015	0.015	0.015
Group A streptococci	0.03–0.12	0.12	0.12
Group B streptococci	0.25–1	1	1
<i>Streptococcus pneumoniae</i>	0.06–32	0.12	0.12
Viridans streptococci	0.03–8	0.06	0.12
<i>Enterococcus faecalis</i>	1–8	2	8
<i>Haemophilus influenzae</i>	0.5–64	1	1
<i>Neisseria meningitidis</i>	0.015–1	0.03	0.5
<i>Listeria monocytogenes</i>	≤0.12–0.25	≤0.12	0.25
<i>Escherichia coli</i>	8–16	8	16
<i>Klebsiella pneumoniae</i>	16–32	32	32
<i>Proteus mirabilis</i>	4–8	4	8
<i>Enterobacter cloacae</i> complex	16–64	64	64
<i>Acinetobacter</i> spp.	4–16	8	8
<i>Clostridioides difficile</i> (formerly <i>Clostridium difficile</i>)	0.10 to >100	1.56	>100
<i>Pseudomonas aeruginosa</i>	32 to >64	32	64

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<i>Acinetobacter</i> spp.	4–16	8	8
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<i>Pseudomonas aeruginosa</i>	32 to >64	32	64
<i>Mycobacterium tuberculosis</i>	0.06–0.25	0.25	0.25
<i>Mycobacterium avium-intracellulare</i>	0.78 to >100	6.25	50
<i>Mycobacterium kansasii</i>	0.025–3.13	0.2	3.13
<i>Mycobacterium marinum</i>	0.1–0.39	0.2	0.39
<i>Mycobacterium fortuitum</i>	12.5–100	50	100
<i>Mycobacterium scrofulaceum</i>	0.1–6.25	0.78	6.25
<i>Legionella</i> spp.	0.001–0.5	≤0.25 ^b	0.125
<i>Bartonella</i> spp.	0.03–0.06	0.12	0.25
<i>Chlamydia</i> spp.	0.0075–0.03	0.015	0.031
<i>Brucella</i> spp.	0.5–1	2	4
<i>Helicobacter pylori</i>	0.125–0.75	0.125	0.125

MIC, Minimal inhibitory concentration; MIC₅₀, minimal inhibitory concentration for 50% of isolates; MIC₉₀, minimal inhibitory concentration for 90% of isolates.

^a MIC ≤2 µg/mL is considered susceptible.

^b Geometric mean MIC.

Rifapentine

Pharmacokinetics

Rifapentine, a cyclopentyl rifamycin, is a more potent and longer-acting rifamycin. ⁵³ It is recommended to be taken with a meal, as food increases the absorption, with a high-fat meal resulting in a 50% increase in serum concentration. Rifapentine achieves high intracellular concentrations, exceeding that of rifampin. CSF concentrations are undetectable. ⁵³ Less than 10% of rifapentine is excreted in urine unchanged. ²⁷ Rifapentine does not induce its own metabolism. The primary route of metabolism, mediated by an esterase enzyme, is nonoxidative to the 25-desacetyl metabolite. Rifapentine also induces CYP3A4 and CYP2C and can interact with the same drugs as rifampin. The degree of interaction correlates with the dose and frequency of administration. Rifapentine is not recommended for use in combination with the HIV protease inhibitors and may be associated with decreased absorption. ²⁷

Adverse Effects

10% of rifapentine is excreted in urine unchanged. ²⁷ Rifapentine does not induce its own metabolism. The primary route of metabolism, mediated by an esterase enzyme, is nonoxidative to the 25-desacetyl metabolite. Rifapentine also induces CYP3A4 and CYP2C and can interact with the same drugs as rifampin. The degree of interaction correlates with the dose and frequency of administration. Rifapentine is not recommended for use in combination with the HIV protease inhibitors and may be associated with decreased absorption. ²⁷

Adverse Effects

Rifapentine is better tolerated than rifampin, likely due to higher protein binding. Once-weekly dosing has a lower incidence of flu-like syndrome ⁵³; however, cases with hemolysis and renal failure have been reported. The incidence of hepatotoxicity is similar to rifampin. Hyperuricemia has been reported with rifapentine in up to 21.3% of patients. ⁵³

Antimicrobial Activity

Rifapentine is active against *M. tuberculosis* only.

Rifaximin

Pharmacokinetics

Rifaximin has an additional ring linking the C-3 and C-4 positions. Absorption after an oral dose is minimal secondary to low intestinal permeability and water solubility. Less than 0.01% of the dose is detectable in plasma, and 97% is recovered unchanged in the stool. Fecal concentrations were 8000 µg/mL in one study. ⁵⁴ The AUC increases 10 to 20 times, and the C_{max} increases 6 to 10 times in patients with cirrhosis. ⁵⁵ Rifaximin can induce cytochrome CYP3A4 but has no significant drug interactions secondary to extremely low systemic concentrations. Although caution is recommended when using rifaximin in patients with Child-Pugh class C cirrhosis, no dosage adjustments are required.

Adverse Effects

Rifaximin is associated with few adverse effects. An unusual case of rifaximin-associated neutropenia was described in a patient receiving 1200 mg daily with severe ulcerative colitis and a transjugular intrahepatic portosystemic shunt, suggesting increased absorption and decreased metabolism. ⁵⁶

Antimicrobial Activity

Rifaximin is bactericidal against enterotoxigenic and enteroaggregative strains of *Escherichia coli*, *Salmonella*, *Shigella*, *Campylobacter* spp. and *C. difficile*. Antiprotozoal activity includes *Cryptosporidium parvum* and *Blastocystis hominis*. ⁵⁷ Use in inflammatory bowel diseases, hepatic encephalopathy, and effects on the microbiome are discussed later.

Immune-Modulating Effects of Rifampin

In addition to bactericidal activity, rifampin has immune-modulating properties. An in vitro study showed inhibition of interleukin (IL)-1β and tumor necrosis factor-α (TNF-α), with increases in IL-6 and IL-10 when rifampin was added to stimulated human monocytes. ⁵⁸ Incubation of rifampin with proinflammatory cytokines in alveolar cells and human epithelial liver cells (HepG2) led to

Cryptosporidium parvum and *Blastocystis hominis*.⁵⁷ Use in inflammatory bowel diseases, hepatic encephalopathy, and effects on the microbiome are discussed later.

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Rifamycins for the Treatment of Tuberculosis

The treatment of active TB requires combination chemotherapy to avoid the selection of naturally occurring drug-resistant mutants. Since the introduction of rifampin five decades ago, the rifamycins have been cornerstone agents in the treatment of TB, particularly when it was shown that rifampin in combination therapy reduced the overall treatment time from 18 to 9 months.⁶⁵ In addition, it has been shown that infections caused by rifampin-resistant strains of *M. tuberculosis* require a longer duration of treatment and are associated with higher rates of treatment failure than those caused by drug-susceptible strains.⁶⁵

The current rifamycins used in the treatment of TB include rifampin, rifapentine, and rifabutin. Rifampin has better penetration into cavitory lung lesions, it has a long track record of safety, it is inexpensive and globally available, and thus it is currently the first-line rifamycin drug for the treatment of TB.⁶⁶ Rifampin is bactericidal against *M. tuberculosis* and, among the first-line agents against TB, has the most potent sterilizing activity (defined as the ability of a drug to kill persistent Page 357 and/or metabolically dormant organisms). This may be in part due to its rapid onset of action.⁶⁷ International guidelines recommend dosing at 10 mg/kg for daily and intermittent treatment with a maximum daily dose of 600 mg. These recommendations, however, were based on historical pharmacokinetic,^{68, 69} toxicity,^{70, 71} and cost arguments⁷² and have largely been refuted by more recent scientific data.⁷³ Data from in vitro systems and animal models provide compelling evidence that rifampin's activity against *M. tuberculosis* is concentration dependent and correlates best with the quotient of the AUC and the MIC.⁷³ Using the mouse model, Jayaram and coworkers⁷⁴ found that the 10 mg/kg dose of rifampin used in humans falls at the low end of a steep and tall dose-response curve, indicating that increasing

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The primary toxicities attributed to rifampin are discussed in detail in the section on adverse reactions and are usually associated with intermittent dosing and are not expected to occur more frequently with increased daily dosing. Transaminitis and increased bilirubin levels after starting treatment do not appear dose related, ⁸⁰ and hepatotoxicity usually occurs with preexisting liver disease or with the use of concomitant hepatotoxins. ⁸¹ Systemic side effects, such as flulike syndrome, are related to intermittent treatment rather than to the actual dose of rifampin. ⁸² Furthermore, rifampin has been used at higher doses for non-mycobacterial infections, such as brucellosis, ⁸³ staphylococcal infections, ⁸⁴ and cutaneous leishmaniasis, ⁸⁵ without reports of increased adverse effects, supporting the idea that the majority of rifampin's adverse effects are idiosyncratic and not dose related.

A recent study has suggested that high-dose rifampin may play a role in treating infections caused by low-level resistant strains ⁸⁶ and may help prevent the emergence of drug resistance. ⁸⁷ Recent data also suggest that regimens containing higher doses of rifampin are safe and could be associated with a survival benefit in patients with severe disease. ⁸⁸

In summary, higher doses of rifampin may help reduce the duration of treatment of TB without a significant increase in adverse events. More extensive, dose-ranging, tolerability, and extended bactericidal activity studies of high doses of rifampin are still undergoing, and their results are expected soon.

Rifapentine, with its more potent in vitro activity (MIC 0.06 vs. 0.25 µg/mL) ⁸⁹ and longer serum half-life (11–18 hours vs. 2–4 hours) than rifampin, ^{27, 90} was developed with the goal to provide an option for once-weekly therapy for TB. ⁹¹ The approved US Food and Drug Administration (FDA) dose for the treatment of active TB is 600 mg administered once or twice

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Recent work using the mouse model of antituberculosis chemotherapy has shown that regimens containing rifapentine administered 5 days per week can achieve durable cure without relapse after 3 months. Their results also indicated that rifapentine is four times more potent than rifampin in the murine model. ⁹⁵ These encouraging results prompted a randomized, multicenter, phase II clinical trial conducted by the Tuberculosis Trials Consortium (TBTC study 29) to determine the safety, tolerability, and antimicrobial activity of a regimen in which rifampin 10 mg/kg was replaced by rifapentine 10 mg/kg administered 5 days per week during the first 8 weeks (intensive phase) of combination treatment of pulmonary TB. ⁹⁶ Their results showed that although the rifapentine regimen was safe, it was not significantly more active than a standard rifampin regimen by the surrogate end point of culture status at completion of the intensive phase. One limitation of this study, however, was administration without food, which has been shown to increase the C_{max} of rifampin ⁹⁷ but decrease rifapentine bioavailability. ⁹⁸

The optimal rifapentine dose and schedule for treating TB in humans have not been established, but the mouse model suggests that the bactericidal and sterilizing activity of rifapentine increases virtually without plateau up to doses of 160 mg/kg and that daily dosing improves outcomes. Dooley and colleagues ⁹⁹ conducted a rifapentine dose-escalation study in healthy adults and showed that daily rifapentine doses of up to 20 mg/kg administered with food were safe and produced rifapentine exposures exceeding those in mice receiving 10 mg/kg. Their results provided safety and tolerability data to support the need of future clinical trials of rifapentine at high daily doses for treatment of TB. A recent trial showed that high rifapentine exposures were associated with high levels of sputum sterilization at completion of an intensive phase, suggesting that regimens that deliver high rifapentine exposures can shorten the duration of treatment to less than 6 months. ¹⁰⁰

Rifabutin has a much longer half-life than rifampin (35 hours vs. 2–4 hours). In clinical trials, the efficacy of rifabutin administered daily or intermittently has been shown to be similar to that of rifampin. ^{101, 102} Rifabutin's primary advantage in the treatment of TB is its reduced induction of hepatic metabolism (cytochrome P450 system) ¹⁰³ and its usefulness in patients with TB and HIV coinfection who are also receiving antiretroviral therapy (ART). Rifampin markedly decreases serum concentrations of the HIV type 1 (HIV-1) protease inhibitors ¹⁰⁴ and has substantial effects

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The most recent guidelines for the treatment of active TB in patients coinfecting with HIV recommend that, despite pharmacokinetic drug interactions, a rifamycin should be included in regimens for patients receiving antiretroviral agents, with dosage adjustment if necessary. If a protease inhibitor-based regimen is used, rifabutin is the preferred rifamycin, but the dose should be decreased from 300 mg daily to 150 mg daily if given with a ritonavir-boosted protease inhibitor. If an efavirenz-based regimen is used, the rifabutin dose should be increased from 300 mg to 450 to 600 mg daily, but the dose of efavirenz remains unchanged. ³¹ Importantly, rifampin remains the rifamycin of choice for patients taking efavirenz-based ART because efavirenz reduces the concentration of coadministered rifabutin.

Recent data examined the tolerability of rifabutin when coadministered with other medications affected by rifampin's induction of Page 358 cytochrome P450, such as methadone and coumadin, and in patients who previously had a rifampin-related adverse effect. Their results showed that rifabutin is well tolerated and has limited the need for dosing adjustments of other medications. Their results also suggest that rifabutin could be considered as an alternative agent in patients who develop rifampin-related liver injury but that rifabutin should be used with caution in patients with other types of underlying liver disease. ¹⁰⁹ Data from transplant patients with TB receiving tacrolimus or cyclosporine also suggest that rifabutin should be used in this population, instead of rifampin, to avoid pharmacologic interactions and achieve a lower risk for acute graft rejection. ¹¹⁰

Rifamycins for Treatment of Latent Tuberculosis Infection

The current standard regimen for the treatment of latent *M. tuberculosis* infection (LTBI) is 9 months of daily isoniazid. Shorter LTBI regimens based on other regimens are growing in

caution in patients with other types of underlying liver disease. ¹⁰⁹ Data from transplant patients with TB receiving tacrolimus or cyclosporine also suggest that rifabutin should be used in this population, instead of rifampin, to avoid pharmacologic interactions and achieve a lower risk for acute graft rejection. ¹¹⁰

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The current standard regimen for the treatment of latent *M. tuberculosis* infection (LTBI) is 9 months of daily isoniazid. Shorter LTBI regimens based on other regimens are growing in popularity, given concerns for isoniazid-related hepatotoxicity, poor adherence attributed to long-duration isoniazid regimens, and the growing proportion of TB cases that are resistant to isoniazid among persons born in areas with high prevalence of the disease.

Early experiments in a mouse model demonstrated the potential efficacy of three short-course regimens that included rifampin, with or without companion medications. The results suggested that 2 months of therapy with rifampin with pyrazinamide or 3 months of therapy with rifampin alone were more effective than 6 months of therapy with isoniazid alone. ¹¹¹ This study prompted a number of clinical trials with three different rifampin-containing regimens. A 2-month regimen of rifampin and pyrazinamide was shown to be as effective as isoniazid. ^{112, 113} Unfortunately, this regimen has been largely abandoned, owing to severe liver toxicity. ¹¹⁴ Results from several randomized trials evaluating the combination of isoniazid and rifampin given for 3 to 4 months have shown an equivalent completion rate, toxicity, and effectiveness as 6 to 9 months of isoniazid. ^{115, 116} A 4-month daily rifampin regimen has been recommended by the American Thoracic Society (ATS) as an alternative regimen in the treatment of LTBI after a few studies consistently found significantly better completion rates and significantly lower serious adverse effects, particularly hepatotoxicity, when compared with 9 months of isoniazid. ¹¹⁷

With the use of animal models, it was shown that once-weekly isoniazid plus rifapentine combination therapy for 18 weeks was an effective preventive regimen, with sterilizing potency and bacillary load reduction comparable with daily isoniazid therapy for 18 weeks. ^{118, 119} These promising results were followed by a large-scale randomized clinical trial that was recently completed and published. Sterling and colleagues ¹²⁰ showed that directly observed, once-weekly therapy with rifapentine plus isoniazid for 3 months was as effective as self-administered daily isoniazid for 9 months, with higher completion rates and lower rates of severe adverse events. This regimen has now been recommended as an equal alternative to the 9-month isoniazid regimen for otherwise healthy patients aged 12 years or older with LTBI. ¹²¹ A recent systematic review compared the efficacy and completion rate of different LTBI drug regimens: isoniazid plus rifapentine once weekly for 12 weeks, 3 to 4 months of daily isoniazid and rifampin, 6 and 9 months of daily isoniazid, and 4 months of rifampin alone. The review showed no significant difference between the groups in terms of efficacy, but regimens of 3 to 4 months' duration were more likely to be completed than longer regimens. ¹²²

Treatment of LTBI in patients with HIV infection continues to be hindered by poor adherence to long-duration regimens, concern about drug resistance, and drug-drug interactions with antiretroviral agents. A recent randomized, open-label, phase III noninferiority trial compared the efficacy and safety of a 1-month regimen of daily rifapentine plus isoniazid with 9 months of isoniazid alone in patients with HIV who were living in areas of high TB prevalence or who had

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Rifamycins for Nontuberculous Mycobacterial Infections

Mycobacterium leprae

Rifampin is rapidly bactericidal against *M. leprae*, with bacteriologic success within 3 weeks of starting therapy. Even a single high dose was efficacious in a mouse footpad model. ¹²⁴ Rifampin should never be used as monotherapy because of rapid development of resistance. Although not FDA approved for leprosy, rifampin is recommended by the WHO for every manifestation of disease. Recommendations are a combination therapy of rifampin with dapsone for paucibacillary disease or rifampin with dapsone and clofazimine for multibacillary disease. The recommended duration of therapy is 6 months for patients with paucibacillary disease and 12 months for those with multibacillary disease, although longer treatment might be required in some patients with a high bacterial burden at diagnosis to prevent relapse. ¹²⁵ Of concern is the emergence of rifampin-resistant strains of *M. leprae* from northern and eastern India, discovered in new cases of leprosy. DNA sequencing showed mutations at previously reported sites in addition to new codon positions, including Ala411 and Gln442, in the *rpoB* gene. The latter mutation leads to upregulation in the messenger RNA expression of the *rpoB* gene, which may lead to overexpression of this gene and other associated genes. ^{126, 127} There is an urgent need to continue monitoring for rifampin resistance in these areas and include new drugs in treatment regimens. The use of rifamycins as chemoprophylaxis against leprosy has been suggested. The overall incidence of leprosy among contacts of patients with newly diagnosed disease can be reduced by a single dose of rifampin. Whether to use this approach more widely is under discussion, and a longer observation time is necessary to demonstrate whether the effect of rifampin prophylaxis will be sustained over long periods. ¹²⁸

Mycobacterium avium-intracellulare Complex

A variety of compounds and their combinations have been reported to be effective against MAC, the rifamycins (particularly rifampin and rifabutin) among them. ¹²⁹ Medical treatment of MAC pulmonary disease has yielded inconsistent results and limited cure rates. Major limitations for effective therapy include the lower bactericidal activity of the rifamycins against MAC than against *M. tuberculosis*, ⁸⁹ the absence of correlation between in vitro activity of rifampin and clinical response for MAC disease, and the markedly decreased immunologic status of the host. ¹³⁰ The limited success of current treatment regimens is also caused by an incomplete understanding of the relationships between the doses of the drugs used, their pharmacokinetics,

effective therapy include the lower bactericidal activity of the rifamycins against MAC than against *M. tuberculosis*,⁸⁹ the absence of correlation between in vitro activity of rifampin and clinical response for MAC disease, and the markedly decreased immunologic status of the host.¹³⁰ The limited success of current treatment regimens is also caused by an incomplete understanding of the relationships between the doses of the drugs used, their pharmacokinetics, and the eventual in vivo treatment outcome. The interactions between rifamycins and macrolides are likely most important, because these drugs are cornerstones of recommended treatment regimens for MAC lung disease. The ATS and the Infectious Diseases Society of America (IDSA) currently recommend for most patients with nodular/bronchiectatic disease a three-times-weekly regimen of clarithromycin (1000 mg) or azithromycin (500 mg), rifampin (600 mg), and ethambutol (25 mg/kg). For patients with fibrocavitary MAC lung disease or severe nodular/bronchiectatic disease, a daily regimen of clarithromycin (500–1000 mg) or azithromycin (250 mg), rifampin (600 mg) or rifabutin (150–300 mg), and ethambutol (15 mg/kg), with consideration of three times per week amikacin or streptomycin early in therapy, is recommended.¹³¹ A recent study showed that currently recommended regimens for MAC lung disease yield important pharmacologic interactions and low concentrations of key drugs.¹³² Trials of new drugs and new strategies are needed.

Disseminated MAC infection occurs largely in patients with HIV infection, especially with CD4⁺ counts at less than 50 cells/ μ L.⁴ The role of rifamycins in the treatment of disseminated MAC disease is unclear. If a rifamycin is used as a second-line agent, most experts would use rifabutin, which is more potent than rifampin against MAC, has a postantibiotic effect of 26 hours at the MIC, and has activity against some strains that are resistant to rifampin. Rifabutin accumulates within phagocytes to a greater extent compared with rifampin¹³³ and is easier Page 359 to use with most antiretroviral agents, especially protease inhibitors. A study of 160 patients with acquired immunodeficiency syndrome (AIDS) and MAC bacteremia compared triple therapy with rifabutin, clarithromycin, and ethambutol with double therapy with clarithromycin with rifabutin or ethambutol. After 12 weeks, the microbiologic response was the same, but patients in the triple-therapy group had increased survival and fewer relapses.¹³⁴ Based on this study, some experts recommend the addition of rifabutin as a third drug for disseminated MAC infection in AIDS, especially with a high mycobacterial burden, but with careful monitoring for drug-drug interactions.¹³⁵ There is no role for rifabutin therapy in the prophylaxis of MAC due to the added toxicity seen when added to a macrolide.

Mycobacterium kansasii

There have been no randomized trials of treatment of disease caused by *M. kansasii*. However, there are several retrospective and prospective studies of various treatment regimens. Strains of *M. kansasii* often are susceptible to rifampin. With rifampin-containing multidrug regimens, sputum conversion within the first 4 months of treatment was achieved in 100% of the cases, and treatment failures and long-term relapses occur in approximately 1% of the cases when treated for at least 12 months.^{136, 137} The ATS and IDSA currently recommend a regimen for treating pulmonary *M. kansasii* disease of rifampin (600 mg/day), isoniazid (300 mg/day), and ethambutol (15 mg/kg/day) for a duration that includes 12 months of negative sputum cultures.¹³¹ Nevertheless, in a recent study, a 12-month course of treatment, although effective in most cases,

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Other Nontuberculous Mycobacteria

The rifamycins show in vitro activity against several other nontuberculous mycobacteria, including *M. marinum*, *M. xenopi*, *M. scrofulaceum*, *M. haemophilum*, *M. terrae* complex, *M. malmoense*, *M. szulgai*, and *M. goodii*. Rifampin is included in the multidrug therapy for infections caused by these nontuberculous mycobacteria. Rapidly growing mycobacteria, such as *M. fortuitum* and *M. chelonae*, are naturally resistant to rifamycins. ⁴³

Rifamycins for Nonmycobacterial Infections

Staphylococcal Infections

Rifampin is a logical treatment choice for staphylococcal infections because of its potent bactericidal activity in vitro against both *S. aureus* and coagulase-negative staphylococci (CoNS). The ability of rifampin to sterilize phagocytes containing bacteria was demonstrated by Mandell and colleagues ¹³⁹ in the 1970s. Rifampin monotherapy was short lived due to the rapid emergence of resistance, especially with high bacterial burdens seen in abscesses and endocarditis, resulting in combination regimens containing rifampin.

In vitro studies, including time-kill and checkerboard studies, were inconsistent when rifampin was added to β -lactams, vancomycin, quinolones, linezolid, and daptomycin and did not correlate with in vivo results. When added to bactericidal antibiotics, there was decreased or unchanged killing, whereas if added to a bacteriostatic antibiotic, there was some increased bactericidal activity. ³ The rabbit model of *S. aureus* endocarditis ¹⁴⁰ provided the first correlation to human infection. The combination of penicillin with rifampin was less effective than penicillin with gentamicin and led to emergence of rifampin-resistant strains. ¹⁴⁰ In a mouse model of MSSA endocarditis, rifampin with nafcillin resulted in an improved survival rate of 77% without selection of resistance. ¹⁴¹ Animal studies of MRSA-related bacteremia and endocarditis yielded variable results. Only daptomycin and linezolid were effective in animal models of MSSA and MRSA endocarditis in reducing colony-forming units in vegetations and septic pulmonary emboli, respectively. ¹⁴² In the 1960s there were reports of successful outcomes when rifampin was used in combination for patients with MSSA and MRSA bacteremia and native valve endocarditis (NVE). ³ Early retrospective data demonstrated emergence of resistance and treatment failure, leading to the conclusion that addition of rifampin to treat *S. aureus* bacteremia without surgery and/or drainage did not improve outcomes and selected for resistance. ^{3, 143} A clinical study of complicated *S. aureus* infections, including NVE, comparing oxacillin or vancomycin with

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A multicenter retrospective study of 357 patients with MSSA bacteremia associated with a deep focus of infection (including endocarditis, pneumonia, septic arthritis, osteomyelitis [OM], abscess, and foreign-body infection) demonstrated that rifampin therapy started within 7 days and continued for at least 14 days from the onset of bacteremia reduced the risk of a fatal outcome by one-third. The positive prognostic impact of adjunctive rifampin in this study merits additional investigation. ¹⁴⁷ A recent meta-analysis that included five randomized controlled trials and two retrospective cohort studies concluded there is insufficient evidence to support the use of rifampin for treatment of *S. aureus* bacteremia (SAB). ¹⁴⁸

Rifampin in combination with other antimicrobials has been studied in *S. epidermidis* NVE. A rabbit model study of methicillin-resistant *S. epidermidis* endocarditis found increased sterilization rates with rifampin-gentamicin-vancomycin and ciprofloxacin plus rifampin. ¹⁴⁹ Human data, however, do not support the routine use of rifampin in NVE caused by CoNS. One exception is *Staphylococcus lugdunensis*, a virulent biofilm-producing species that causes fulminant NVE, pacemaker lead endocarditis, and bone and joint infections and is associated with a high mortality. ¹⁵⁰ Combination therapy may be indicated for bactericidal effects, and rifampin has been used successfully in *S. lugdunensis* infections. ¹⁵⁰

Staphylococcal Biofilms: Foreign-Body Infections and the Role of Rifampin

The ability of staphylococci to produce biofilm significantly affects the chronicity of infections, particularly those with indwelling devices. Biofilm is a community of bacterial cells encased within a polymeric matrix produced by the organisms that adhere to a living or foreign body. ¹⁵¹ Within biofilms organisms exist in distinct populations, in different stages of growth, and exhibit heterogeneous gene expression. Antimicrobial resistance occurs by decreased penetration and tolerance induced by the anoxic environment and nutrient depletion. ¹⁵² Rifampin in vitro is effective against staphylococci in various stages of growth, including metabolically dormant bacteria, penetrates *S. epidermidis* biofilm, and can prevent biofilm formation within minutes. The lipid solubility of rifampin, activity within the acidic environment within some biofilms, and accumulation within neutrophils facilitate the activity of rifampin. ¹⁵³ which is commonly used in

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In vitro the addition of rifampin to *S. epidermidis* and MRSA biofilm infections improved the susceptibility to other antibiotics and decreased the adherence of bacteria to the foreign material.¹⁵⁴ Rifampin with linezolid and daptomycin in animal models of foreign-body infection showed higher activity against planktonic bacteria when rifampin was added.^{155, 156} When used with daptomycin, tigecycline, and linezolid against MRSA biofilms in catheter lock solutions, rifampin combinations were better at eradicating biofilm compared with vancomycin and linezolid alone.¹⁵⁷ New delivery systems of rifampin, including polymethylmethacrylate beads¹⁵⁸ and lipid-based nanoparticles,¹⁵⁹ show promise in vitro in reducing biomass-established biofilms and decreasing viability of intracellular bacteria.

In vivo studies and case reports have used combinations of rifampin with vancomycin, β -lactams, and aminoglycosides for the treatment of *S. epidermidis* prosthetic valve endocarditis (PVE). An early retrospective study of 23 patients with methicillin-resistant *S. epidermidis* PVE found a cure rate in 87% of patients treated with vancomycin with rifampin (900–1200 mg/day) or with rifampin and an aminoglycoside. Only 38% of those receiving a β -lactam with rifampin, with or without Page 360 an aminoglycoside, were cured. The lower response in the patients treated with β -lactams was attributed to heteroresistance within the *S. epidermidis* population.¹⁶⁰ This study and multiple case reports led to inclusion of rifampin as part of a triple therapeutic regimen for CoNS PVE with vancomycin, rifampin, and gentamicin for 2 weeks, followed by vancomycin and rifampin for at least 4 weeks. These recommendations were expanded to include *S. aureus* PVE, for which vancomycin plus rifampin (300 mg every 8 hours) for at least 6 weeks, with gentamicin for the first 2 weeks, is recommended.¹⁶¹

Vancomycin-Intermediate and Vancomycin-Resistant *Staphylococcus aureus* and Rifampin

Since the emergence of vancomycin-intermediate *S. aureus* (VISA) and vancomycin-resistant *S. aureus* (VRSA) strains, alternative antibiotics have been studied in vitro. Rifampin was active against 92% of VRSA isolates but only 51% of VISA isolates in one study.¹⁶² A possible explanation may be found in another study that identified five *rpoB* mutations in 71% of VISA strains.¹⁶³ Among these strains, 55% were resistant to rifampin, supporting a possible link between resistance to vancomycin and rifampin exposure in *S. aureus*. In 29% of VISA strains there was no *rpoB* mutation, suggesting a different resistance mechanism. When rifampin-resistant mutants from nine clinical MRSA isolates were tested, more than 95% had decreased vancomycin susceptibilities. The *rpoB* mutations were observed in 31% of rifampin-susceptible VISA strains, raising the possibility that selection of resistance in these strains occurred by vancomycin and/or other antibiotics.¹⁶³

Prosthetic Joint Infections

Page 362 *Aspergillus* spp., and *Saccharomyces cerevisiae*. In a mouse model, combination therapy with AMB and rifampin was more effective against *H. capsulatum*, *Blastomyces dermatitidis*, and *Aspergillus* spp. An in vitro study on killing of *Candida* biofilms using AMB alone or in combination with either rifampin or clarithromycin found a synergistic effect of rifampin and AMB in 66.6% of *C. parapsilosis*, 42.8% of *C. albicans*, and 33.3% of *C. glabrata* biofilms. ²²⁰ In vitro data show synergy with rifampin and AMB in 68.7% of *Fusarium* spp. ²²¹ Rifampin with AMB significantly enhanced the activity of AMB alone in vitro against 95 clinical isolates of *Fusarium solani* and *Aspergillus flavus* spp. complex from patients with keratomycosis. ²²² There are few clinical reports, with insufficient data to support the use of rifampin in combination with antifungal agents.

Second-Line Indications for Rifamycins

Rifabutin has been successful as a “rescue therapy” for *H. pylori* infection. A combination of rifabutin with amoxicillin and a proton pump inhibitor for 10 days used after failure of standard quadruple therapy was successful and well tolerated in 50% to 66.6% of patients in two studies. ^{223, 224} More recently, this regimen was evaluated in a phase III, randomized, double-blind, controlled study in treatment-naïve *H. pylori*-positive adult patients complaining of epigastric pain/discomfort. The eradication rate was higher with the combination of rifabutin/amoxicillin/omeprazole than with the active comparator amoxicillin/omeprazole (83.3% vs. 57.7%, respectively). ⁵² There is literature supporting rifabutin use at dosages of 150 mg or 300 mg/day as part of regimens for *H. pylori*.

Q fever endocarditis, caused by *Coxiella burnetti*, is usually treated with doxycycline in combination with hydroxychloroquine. Doxycycline with rifampin (900 mg/day) was effective in some case reports, and erythromycin with rifampin was successful in chronic Q fever of pregnancy. ²²⁵ Doxycycline is first-line treatment for scrub typhus, but in areas where doxycycline resistance is present, rifampin is an alternative. ²²⁶ Doxycycline is also the treatment of choice for human granulocytic ehrlichiosis, but rifampin is successful in treating young children. ²²⁷ Rifampin at doses of 300 to 600 mg/day reduces pruritus associated with cholestasis, possibly by induction of microsomal enzymes that promote glucuronidation of toxic bile salts. A review of 12 trials documented a significant decrease in pruritus. ²²⁸

Chemoprophylaxis

Antibiotics for chemoprophylaxis of *N. meningitidis* contacts include rifampin, ciprofloxacin, minocycline, and ceftriaxone. A 2-day course of rifampin (600 mg twice daily) is effective in eradicating the bacteria from 75% to 95% of carriers; however, adverse drug reactions, drug interactions, and selection of resistant isolates have limited use to young children. An extensive review found ceftriaxone to be more effective than rifampin after 1 to 4 weeks of follow-up. Rifampin was effective for up to 4 weeks after treatment, but resistant isolates were seen, raising concern that use of rifampin during an outbreak may lead to the circulation of resistant strains. Use of ciprofloxacin, ceftriaxone, or penicillin should be considered. ²²⁹

The *H. influenzae* type b conjugate vaccines have significantly decreased the incidence of invasive disease in children. Chemoprophylaxis is recommended for close contacts of a child with invasive *H. influenzae* type b infection using rifampin at doses of 20 mg/kg in children and 600 mg in adults daily for 4 days, based on earlier studies showing elimination of oropharyngeal

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Another area of infection prophylaxis is the rifampin-minocycline-impregnated catheter, which has been demonstrated to be effective in decreasing catheter-related bacteremia. [232](#) Two studies in pediatric burn patients [233](#) and in immunocompromised patients with transplants, cancer, or on dialysis [234](#) concluded that these catheters are effective in reducing the incidence of catheter-associated bacteremia. The CATCH trial, comparing minocycline-rifampin-impregnated central venous catheters, heparin-impregnated catheters, and standard catheters in children admitted to pediatric ICUs, found that the antibiotic-impregnated catheter reduced the risk of bloodstream infection by 57% compared with standard catheters and 58% compared with heparin-impregnated catheters. [235](#)

For decolonization of MRSA carriers, mupirocin ointment and chlorhexidine are effective in 60% of patients. Oral therapy is recommended if topical regimens fail, which is seen with mupirocin-resistant isolates. Rifampin for 5 to 10 days in combination with another agent is recommended if the strain is susceptible and other regimens are ineffective. [161](#)

Novel and Future Indications for Rifampin

Rifampin has been successful in the treatment of cholestasis of pregnancy, in combination with ursodeoxycholic acid, in women unresponsive to ursodeoxycholic acid alone. [236](#) The mechanism is thought to be due to enhancement of bile acid detoxification, bilirubin conjugation, and bilirubin excretion. A 6- to 9-month course of rifampin was effective treatment in 30 patients with idiopathic granulomatous lobular mastitis, a chronic inflammatory condition of unclear etiology associated with masses, abscesses, and the sinus tracts. [237](#) A recent study in Ghana confirmed the effectiveness of an 8-week course of rifampin with streptomycin in curing Buruli ulcer caused by *Mycobacterium ulcerans*. [238](#)

Two innovative studies of rifampin looked at suppression of osteolysis after joint replacement and the prevention of neurodegenerative diseases. Using in vitro data and a mouse calvarial osteolysis model, rifampin inhibited titanium-induced osteolysis and osteoclastogenesis in vivo and reduced expression of osteoclast-specific markers and osteoclast formation through regulation of receptor activator of nuclear factor- κ B ligand signaling. [239](#) Using a transgenic mouse model of Alzheimer disease, rifampin for 1 month reduced accumulation of amyloid- β oligomers, tau hyperphosphorylation, synapse loss, and microglial activation. The 1 mg/day dose improved memory similar to the nontransgenic littermates. [240](#) Future studies are needed to validate a role for rifampin in the prevention of periimplant osteolysis, diseases with excessive osteoclast activity, and in neurodegenerative diseases.

Over the last decade, numerous observational studies have been conducted to evaluate the effect of an antibiotic envelope that elutes two antibiotics, rifampin and minocycline, on rates of cardiac implantable electronic device (CIED) infections and have yielded variable results. [241](#), [242](#) More recently, a randomized, controlled clinical trial showed that adjunctive use of an antibacterial envelope resulted in a significantly lower incidence of major CIED infections than standard-of-care infection prevention strategies alone (12-month Kaplan-Meier estimated event rate, 0.7% and 1.2%, respectively; hazard ratio, 0.60), without a higher incidence of

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Rifaximin: A Selective Gastrointestinal Rifamycin

Rifaximin is a unique rifamycin with negligible systemic absorption, enhanced fecal concentrations, decreased systemic toxicity, and few drug interactions. Rifaximin is bactericidal against many enteric pathogens, including enterotoxigenic and enteroaggregative *E. coli*, *Salmonella*, *Shigella*, and *Campylobacter*, ⁵⁷ and is active against *C. difficile*, although high-level resistance has been reported. ²⁴⁴ Rifaximin has antiprotozoal activity, including against *C. parvum* and *B. hominis*. ⁵⁷ In addition to its role in traditional single-pathogen-based antibiotic activity, rifaximin affects the microbiome of the GI tract. Rifaximin can downregulate inflammation by inhibiting activation of nuclear factor- κ B and reducing expression of the proinflammatory cytokines IL-1 and TNF- α . ²⁴⁵ Rifaximin may also modulate inflammatory and immune responses in gut epithelial cells. ²⁴⁶ Clinical trials of traveler's diarrhea demonstrate that rifaximin (400 mg twice daily) was similar to traditional antimicrobial agents in decreasing time to symptom resolution. ⁵⁷ Rifaximin was 72% effective in chemoprophylaxis of traveler's diarrhea in Mexico, where most cases are caused by *E. coli*. ²⁴⁷ At subinhibitory concentrations, rifaximin reduced expression of bacterial enterotoxins and surface adhesion intestinal-binding factors of enterotoxigenic *E. coli*. ²⁴⁸ However, the fluoroquinolones are preferred for chemoprophylaxis due to an effectiveness of over 90% and activity against most invasive intestinal pathogens. ²⁴⁹ Rifaximin (550 mg twice daily) is approved for the prevention of hepatic encephalopathy.

Page 363 In one study rifaximin use led to fewer breakthrough episodes of hepatic encephalopathy and significantly reduced the risk for hospitalization compared with placebo. ²⁵⁰ The reduction of microbes that degrade nitrogenous compounds results in decreased ammonia levels and was more than or equally effective as lactulose and other agents and better tolerated. ²⁵¹

Small trials of rifaximin for *C. difficile* infection have been published, although rifaximin may best be used for recurrent *C. difficile* infection, with a response rate as high as 83% in one study. ²⁵² A 2-week course of rifaximin is often used after a 6-week course of vancomycin for recurrent disease. There is concern about the development of resistance, however. Using the rifampin Etest, one study found 14 of 80 clinical isolates had an MIC greater than 32 μ g/mL. Of these 14 isolates, 9 (64%) belonged to the epidemic BI/NAP1/027 group. ²⁵³ In small bowel overgrowth, rifaximin may play a role in suppressing microbial colonization. ²³³ Rifaximin is FDA approved for treatment of irritable bowel disease without constipation. Patients reported a modest improvement in bloating, abdominal pain, and stool consistency. Rifaximin may be beneficial in inflammatory bowel diseases, including ulcerative colitis, Crohn disease, and pouchitis, as a steroid-sparing regimen, ²⁴⁴ possibly due to the effect on both the microbiome and the (subsequent) antiinflammatory effect. Studies of rifaximin alone or in combination with other therapies have demonstrated improved rates of remission and/or clinical improvement in Crohn disease, ulcerative colitis, and pouchitis. ²⁵⁴ There may be a role for rifaximin in the treatment of

the (subsequent) antimicrobial effect. Studies of rifaximin alone or in combination with other therapies have demonstrated improved rates of remission and/or clinical improvement in Crohn disease, ulcerative colitis, and pouchitis. ²⁵⁴ There may be a role for rifaximin in the treatment of diverticular disease when used 7 days per month along with traditional therapies. ²⁵⁵ Caution is warranted for such empirical therapies because rifampin-resistant strains of *S. aureus* and CoNS were found in the perineum, hands, and upper extremities after rifaximin therapy that persisted up to 9 weeks after discontinuation, consistent with other studies demonstrating rifaximin-resistant *E. coli* in the intestinal tract. ^{256, 257} In a recent study, a delayed-release formulation of rifaximin had a protective effect in reducing the number and severity of small bowel mucosal lesions and large erosions and/or ulcers when used with a nonsteroidal anti-inflammatory drug. This effect is believed to be secondary to a shift in the microbiota away from proinflammatory gram-negative bacteria. ²⁵⁸

Rifamycins in Development

New rifamycins include aerosolized formulations, rifamycin-loaded scaffolds for prevention of orthopedic prosthetic infections, and compounds that circumvent development of resistance. Advantages of aerosol formulations are decreased dosing, improved compliance, decreased adverse events, and improved pharmacodynamics, because particles attain high intracellular concentrations in pulmonary macrophages. Two investigational compounds are a dry powder formulation with rifampin-derived microcrystals coated with polymers ²⁵⁹ and liposomes coated with chitosan-xanthan gum. ²⁶⁰ A new inhaled formulation of rifampin with colistin, aimed at delivering both drugs directly to the respiratory tract, was found to have synergistic activity, high aerosolization efficiency, and moisture protection, offering a less toxic formulation for resistant gram-negative respiratory infection. ²⁶¹ Synthetic bone scaffolds constructed by electrospinning nanofiber poly(caprolactone) with concentrations of 10% and 20% rifampin, when inoculated with *S. epidermidis* and *P. aeruginosa*, completely inhibited bacterial growth and prevented biofilm formation within the scaffolds through the first 6 hours. ²⁶²

Rifamycin SV MMX (RIF-MMX) is a semisynthetic derivative of rifamycin SV in a nonabsorbable modified-release multimatrix structure that delivers the active ingredient directly into the colon. ²⁶³ The bioavailability is less than 0.1%, and close to 90% of the dose is recovered in feces. RIF-MMX is under investigation for treatment of traveler's diarrhea, *C. difficile* infection, diverticulitis, inflammatory bowel diseases, and hepatic encephalopathy. A 3-day course of RIF-MMX versus placebo was well tolerated and effective in achieving clinical cure within 120 hours in patients with traveler's diarrhea (81.4% of RIF-MMX vs. 56.9% of placebo). There was decreased susceptibility in bacterial isolates remaining after therapy, but this was not associated with diminished efficacy, suggesting an additional nonbacterial mechanism of action. ²⁶⁴

TNP-2092 is a multitargeting drug conjugate (rifamycin-quinolizone hybrid) that exerts its antibacterial activity by inhibiting three different drug targets: RNA polymerase, DNA gyrase, and topoisomerase IV. Compared with rifaximin, TNP-2092 possesses a similar pharmacokinetic profile and antibacterial spectrum but has a lower frequency of spontaneous resistance due to its multitargeting mechanism of action. In a mouse *C. difficile* infection model, TNP-2092 demonstrated potent efficacy superior to metronidazole and vancomycin, with no relapse observed after treatment. ²⁶⁵ These results support the further development of TNP-2092 as a locally acting drug for the treatment of GI and liver disorders such as traveler's diarrhea, hepatic

profile and antibacterial spectrum but has a lower frequency of spontaneous resistance due to its multitargeting mechanism of action. In a mouse *C. difficile* infection model, TNP-2092 demonstrated potent efficacy superior to metronidazole and vancomycin, with no relapse observed after treatment. ²⁶⁵ These results support the further development of TNP-2092 as a locally acting drug for the treatment of GI and liver disorders such as traveler's diarrhea, hepatic encephalopathy, irritable bowel syndrome, and *C. difficile* diarrhea.

TNP-2198 (a rifamycin-nitroimidazole hybrid) has been designed and evaluated as a novel dual-targeted antibacterial agent for the treatment of microaerophilic and anaerobic bacterial infections. In vitro, TNP-2198 exhibits greater activity than a 1:1 molar mixture of the parent drugs and exhibits activity against strains resistant to both rifamycins and nitroimidazoles. In vivo, TNP-2198 demonstrates potent efficacy in established rodent models of *H. pylori* and *C. difficile* infection. TNP-2198 is currently under clinical development for the treatment of *H. pylori* infection, *C. difficile* infection, and bacterial vaginosis. ²⁶⁶

Studies of rifampin-loaded nanoparticles showed enhanced delivery of rifampin in vitro into the intracellular compartment of polymorphonuclear leukocytes ²⁶⁷ and targeted delivery to alveolar macrophages in vitro and in a rat model. ²⁶⁸ Nanoparticles carrying rifamycins are promising new delivery systems for the treatment of TB, *S. aureus*, and other intracellular bacterial infections.

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Metronidazole

Jerod L. Nagel

David M. Aronoff

Short View Summary

Description

- Metronidazole is a nitroimidazole drug that is widely prescribed as a first-line agent for various anaerobic and parasitic infections and is recommended as an alternative agent (in patients unable to take oral fidaxomicin or vancomycin) for treatment of *Clostridioides difficile* (formerly *Clostridium difficile*) colitis.
- Although metronidazole has its greatest activity against anaerobes (and microaerophils), it exhibits very limited activity against aerobic bacteria.
- Several related nitroimidazole compounds, including tinidazole and secnidazole, are also available for treating infections and may offer advantages over metronidazole (including more convenient dosing and preserved antimicrobial susceptibility).

Spectrum

- There is minimal resistance to metronidazole in *Bacteroides* and other gastrointestinal anaerobes. However, emerging resistance is noted in *Gardnerella vaginalis*, *Helicobacter pylori*, and *C. difficile*, and metronidazole lacks activity against a number of prevalent non-spore-forming, gram-positive anaerobic bacteria including *Actinomyces*, *Bifidobacterium*, *Lactobacillus*, *Propionibacterium*,¹⁻³ and *Cutibacterium acnes* (formerly *Propionibacterium acnes*).

Clinical Pharmacology

- Metronidazole is available in many formulations; it has excellent bioavailability and excellent tissue penetration, including penetration into the central nervous system.

Toxicity

- Most patients tolerate metronidazole, but common adverse effects include nausea, diarrhea, dry mouth, metallic taste, candidal vaginitis, and stomatitis. It has also been reported to cause unpleasant disulfiram-like reactions when co-administered with alcohol. Serious adverse effects include dizziness, headache, confusion, vertigo, insomnia, Stevens-Johnson syndrome, pancreatitis, ophthalmologic toxicity (myopia and blurred vision), ototoxicity, bullous

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History

Metronidazole was first synthesized in the 1950s when the pharmaceutical company Rhône-Poulenc was searching for an effective antitrichomonal drug for the treatment of vaginal trichomoniasis. ⁴ Initially, a crude extract of a *Streptomyces* bacterium was found to kill *Trichomonas vaginalis*, and the active component was determined to be azomycin, a previously characterized nitroimidazole antibiotic. ⁵ Metronidazole (initially called 8823 R.P.) was a synthetic analogue of azomycin that was both more active against *T. vaginalis* and less toxic. ^{4, 6} This discovery soon led to its successful use in human clinical studies of trichomoniasis. ^{7, 8} The presumed antibacterial activity of metronidazole was discovered incidentally in 1962 by Shinn, who reported improvement in ulcerative gingivitis in a patient treated for a concomitant *T. vaginalis* infection, a result he confirmed in a series of patients with ulcerative stomatitis. ⁹

Mechanism of Action

Metronidazole and other nitroimidazoles (e.g., nimorazole, ornidazole, ronidazole, secnidazole, tinidazole) are inert, and their spectrum of antimicrobial activity is determined by the capacity of susceptible organisms to activate the drugs once they enter the cell via passive diffusion (although a receptor for metronidazole uptake has not been described, some studies suggest active mechanisms might govern drug transport into microbial cells). ¹⁰ The structures of metronidazole and related compounds are illustrated in **Fig. 27.1**. Their bactericidal and parasiticidal activities are rapid and proportional to the concentration of the activated drugs within the target cell. ¹¹ The members of this class of antibiotics are therefore best considered as prodrugs, which are activated through a reduction step to form highly reactive products that interact with intracellular targets (see later). ¹² In fact, the active form of metronidazole is rapidly converted back to the prodrug in the presence of oxygen, explaining why the drug is selective for anaerobic or microaerophilic cells. ^{13, 14} Metronidazole's mechanism of action describes that of the other nitroimidazoles.

The nitroimidazoles share a heterocyclic structure consisting of an imidazole-based nucleus with a nitro group, NO₂, in position 5. There are four major steps involved in the mechanism of action of metronidazole that result in the intracellular formation of critical redox intermediate metabolites. ¹⁵ In the first two steps, the drug enters cells by passive diffusion and an electron is transferred to the nitro group of metronidazole, resulting in the production of a short-lived nitroso free radical, which is cytotoxic and can interact with cellular DNA. ¹ This process of